vertically downwards. In order to move the body without acceleration (or with constant velocity), an external applied force $\vec{F}_a$ equal in magnitude but opposite to that of gravitational force $\vec{F}_g$ has to be applied on the body i.e., $\vec{F}_a = -\vec{F}_g$. This implies that $\vec{F}_a = +mg\,\hat{j}$. The positive sign implies that the applied force is in vertically upward direction. Hence, when the body is lifted up its velocity remains unchanged and thus its kinetic energy also remains constant.

The gravitational potential energy (U) at some height h is equal to the amount of work required to take the object from the ground to that height h.

$$U = \int \vec{F}_a . d\vec{r} = \int_0^h \left|\vec{F}_a\right| \left|d\vec{r}\right| \cos\theta \quad (4.17)$$

Since the displacement and the applied force are in the same upward direction, the angle between them, $\theta = 0°$. Hence, $\cos 0^0 = 1$ and $\left|\vec{F}_a\right| = mg$ and $\left|d\vec{r}\right| = dr$.

$$U = mg\int_0^h dr \quad (4.18)$$

$$U = mg\left[r\right]_0^h = mgh \quad (4.19)$$

Note that the potential energy stored in the object is defined through work done by the external force which is positive. Physically this implies that the agency which is applying the external force is transferring the energy to the object which is then stored as potential energy. If the object is allowed to fall from a height h then the stored potential energy is converted into kinetic energy.

Note

- **How can an object move with zero acceleration (constant velocity) when the external force is acting on the object?**

  It is possible when there is another force which acts exactly opposite to the external applied force. They both cancel each other and the resulting net force becomes zero, hence the object moves with zero acceleration.

- **Why should the object be moved at constant velocity when we define potential energy?**

  If the object does not move at constant velocity, then it will have different velocities at the initial and final locations. According to work-kinetic energy theorem, the external force will impart some extra kinetic energy. But we associate potential energy to the forces like gravitational force, spring force and coulomb force. So the external agency should not impart any kinetic energy when the object is taken from initial to final location.

EXAMPLE 4.8

An object of mass 2 kg is taken to a height 5 m from the ground $\left(g = 10\,m\,s^{-2}\right)$.

(a) Calculate the potential energy stored in the object.
(b) Where does this potential energy come from?
(c) What external force must act to bring the mass to that height?
(d) What is the net force that acts on the object while the object is taken to the height 'h'?

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